

Active Workspace Client Customization/Configuration

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Using XML rendering templates (XRT) with Active Workspace

XRT files use the XML format. They are used to configure layout in Teamcenter clients, including Active Workspace, based on object type, group, and role. XRT files are also commonly referred to as style sheets; however, they neither follow CSS or XSL standards, nor do they perform transformations.

In Active Workspace, XRT files control areas such as the secondary work area and the tools and information panel.

Active Workspace style sheet preference names follow this format:

AWC_<type-name>.*RENDERING

Options

Preferences By Search

Options

Search In

☒ Keys ☒ Values

☒ Description ☒ All

Match

☐ Case ☐ Entire Word

Wildcard *

Preferences List

Search On Keywords

AWC_ChangeRe

Filter by category

Filter by protection scope

Name	Location
AWC_ChangeRequest.CREATERENDERING	Site
AWC_ChangeRequestRevision.INFORENDERING	Site
AWC_ChangeRequestRevision.SUMMARYRENDERING	Site

Preferences found: 3

Options | Filters | Search | Organization

Definition | Instances | Category | Import | Export

Click on the "Edit" button to modify the definition and update any field in order for the "Save" button to be enabled. Note that the "Description" field must not be empty. Click on the "Save" button to save the definition of the existing preference.

Name AWC_ChangeRequestRevision.SUMMARYRENDERING Location Site Protection Scope Site

Category Change Management Active Workspace Environment Disabled Type String Multiple Single

Description Registration of summary stylesheet for ChangeRequestRevision objects.

Value Cm1ChangeRequestRevisionSummary

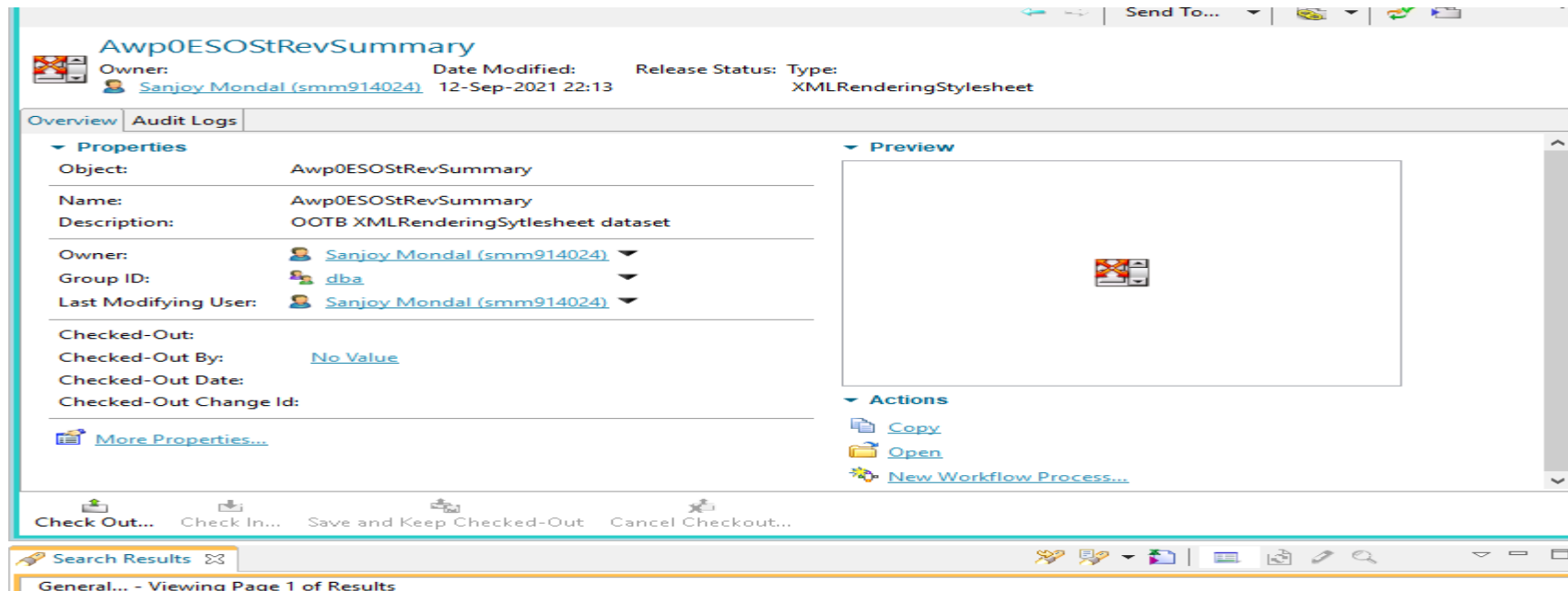
Edit Save

Close

Create Active Workspace-specific style sheets

If you want to have different style sheets in Active Workspace than you have in other Teamcenter clients, you can create AWC_ preferences in the rich client to tell Active Workspace which style sheet to use. This has no effect on the other clients or on any customer-created style sheets

1. Create a dataset with a type of **XMLRenderingStylesheet**



Create Active Workspace-specific style sheets

Step 2. Create a preference in the rich client using the following parameters

AWC_<type-name>.SUMMARYRENDERING
AWC_<type-name>.CREATERENDERING
AWC_<type-name>.INFORENDERING
AWC_<type-name>.REVISERENDERING
AWC_<type-name>.SAVEASRENDERING

The screenshot shows the 'Options' dialog box with the 'Preferences By Search' tab selected. The dialog is divided into several sections:

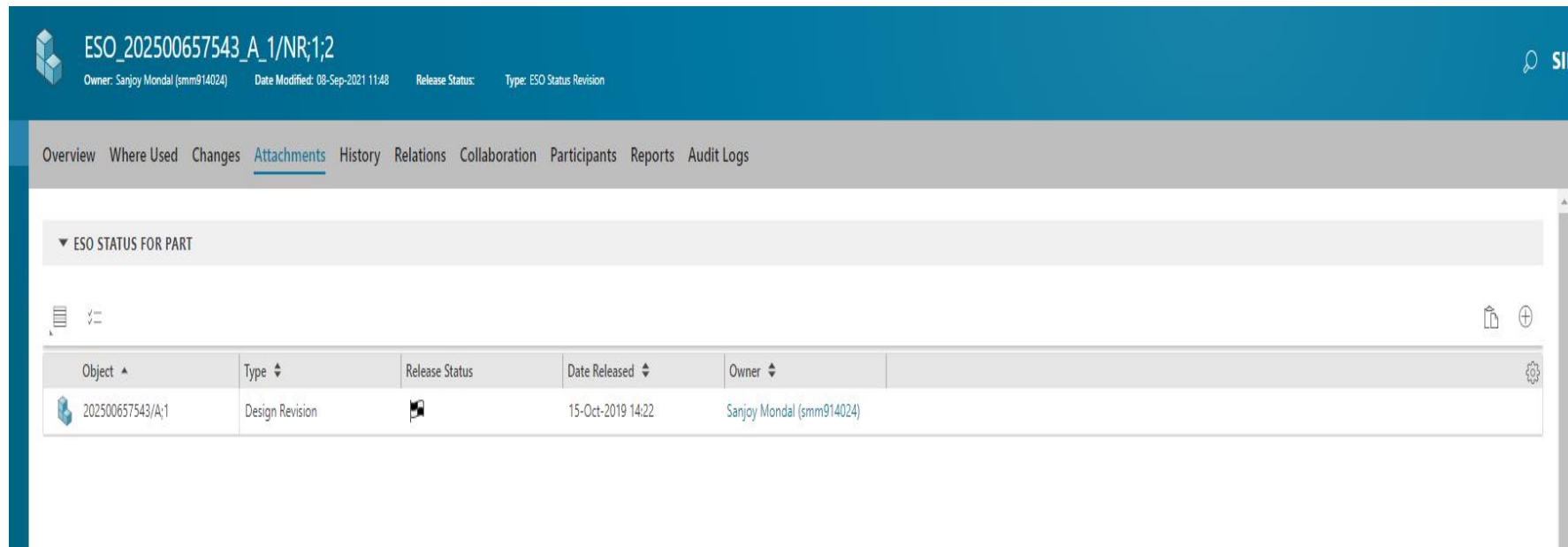
- Options:** Includes checkboxes for 'Search In' (Keys, Values, Description, All) and 'Match' (Case, Entire Word). A 'Wildcard' field is also present.
- Preferences List:** A table showing the search results. The first entry is 'AWC_T5_ESOStRevision.SUMMARYRENDERING' with location 'Site'. Below the table, it says 'Preferences found: 1'.
- Definition:** A section for defining the preference. It includes a text area with instructions: 'Click on the "Edit" button to modify the definition and update any field in order for the "Save" button to be enabled. Note that the "Description" field must not be empty. Click on the "Save" button to save the definition of the existing preference.' Below this are fields for:
 - Name:** AWC_T5_ESOStRevision.SUMMARYRENDERING
 - Location:** Site
 - Protection Scope:** Site
 - Category:** Active Workspace
 - Environment:** Disabled
 - Type:** String
 - Multiple:** Single
 - Description:** AWC_T5_ESOStRevision.SUMMARYRENDERING
 - Value:** Awp0ESOStRevSummary
- Buttons:** 'Edit' and 'Save' buttons are at the bottom right of the definition section. A 'Close' button is at the bottom right of the dialog.

Create Active Workspace-specific style sheets

When rendering style sheets, Active Workspace first searches for **AWC_<type_name>.[SUMMARYRENDERING, CREATERENDERING, ...]**. If no match is found, it searches for **<type-name>.[SUMMARYRENDERING, CREATERENDERING, ...]**. If it still does not find a match, it continues with the standard lookup mechanism for style sheets.

Edit the stylesheet to make your ui changes.

Example- Added a new section in the Navigation Tab Attachment of custom business object as shown below



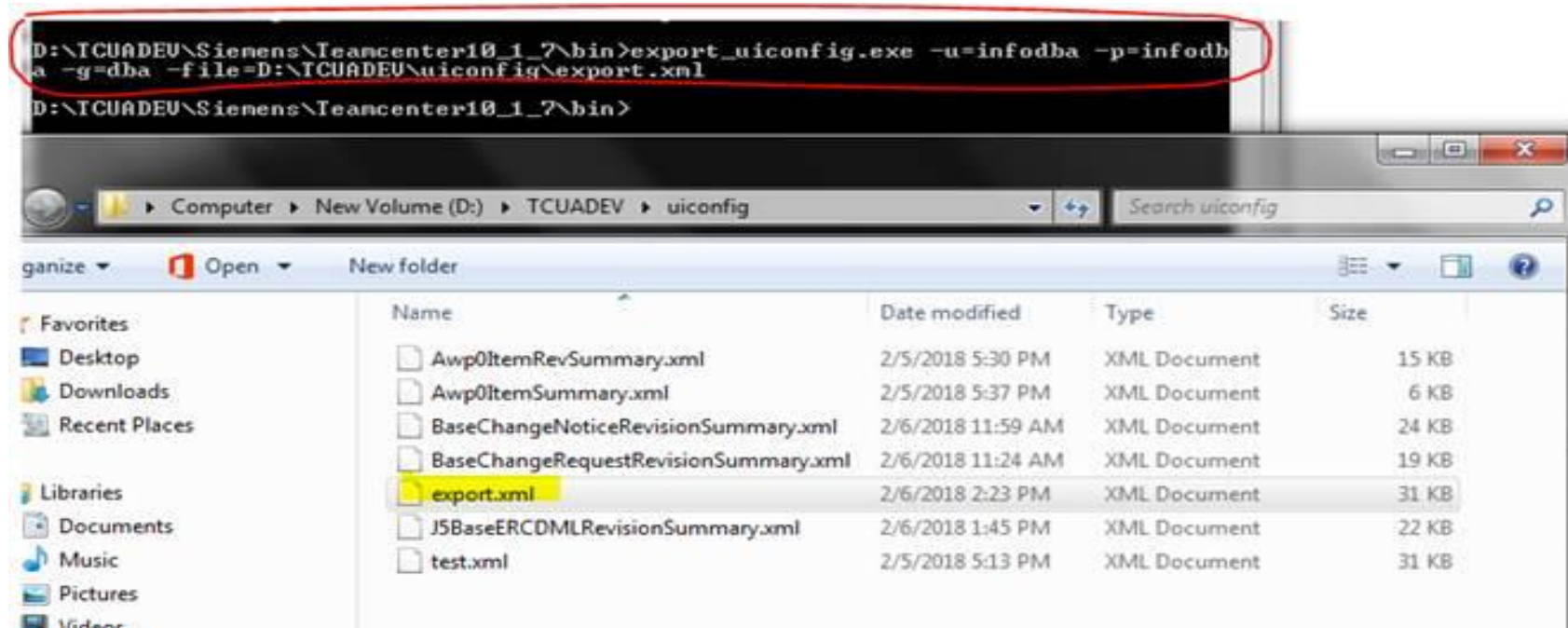
The screenshot shows the ESO_202500657543_A_1/NR;1;2 interface. The top header bar is blue and contains the object ID, owner (Sanjoy Mondal), date modified (08-Sep-2021 11:48), release status, and type (ESO Status Revision). Below the header is a navigation bar with tabs: Overview, Where Used, Changes, Attachments (selected), History, Relations, Collaboration, Participants, Reports, and Audit Logs. The main content area shows a section titled "ESO STATUS FOR PART" with a table of attachments.

Object	Type	Release Status	Date Released	Owner
202500657543/A;1	Design Revision		15-Oct-2019 14:22	Sanjoy Mondal (smm914024)

Configuring tables

- Active Workspace provides two utilities, **export_uiconfig** and **import_uiconfig**, that let administrators specify table column configurations
- 1. Use the utility **export_uiconfig** to generate XML containing Column Configurations used by AWS
 - a) Open command prompt
 - b) Go to %TC_ROOT%\bin
 - c) Set the Environment Variables (execute %TC_DATA%\tc_profilevars.bat)
 - d) Run the **export_uiconfig** utility as below


```
export_uiconfig.exe -u=infodba -p=infodba -g=dba -file=D:\TCUADEV\uiconfig\export.xml
```



Configuring tables

Example

Initially in My Changes looks below. Let add one more column of property say object_type after ID

Sanjoy Mondal (smm914024)-Engineering/Designer-Latest Working

Select: Owner Select: Category

← Changes

All Submitted Saved Approved Disapproved

Advanced Search

4 results found for "All" >

IN PROCESS	ID	REVISION	SYNOPSIS	MATURITY	DATE CREATED	LAST
 False	17JM000009	A	Sanjoy Test	Complete	05-Feb-2018 15:48	06-Fel
 False	17JM000000_02	A	sanjoy task	Complete	05-Feb-2018 15:49	06-Fel
 False	17JM000008	A	Test Sanjoy	Complete	05-Feb-2018 10:35	05-Fel
 False	17JM000000_01	A	Test Task	Complete	05-Feb-2018 10:36	05-Fel

Configuring tables

For this search for fnd0Change in export.xml.

```

] <Client abbreviation="AWClient" name="AWClient">
] <ClientScope hostingClientName="" name="All" uri="fnd0Change">
]   <ColumnConfig columnConfigId="fnd0ChangeColConfig" sortBy="7" sortDirection="Descending">
      <ColumnDef objectType="ChangeItemRevision" propertyName="fnd0InProgress" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="awp0Item_item_id" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="item_revision_id" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="object_name" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="CMMaturity" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="creation_date" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="last_mod_date" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="cm0Requestor" width="200"/>
    </ColumnConfig>
  </ClientScope>
</Client>

```

Configuring tables

Edit the above file and add one more ColumnDef as shown below

```
<Client abbreviation="AWClient" name="AWClient">
  <ClientScope hostingClientName="" name="All" uri="fnd0Change">
    <ColumnConfig columnConfigId="fnd0ChangeColConfig" sortBy="7" sortDirection="Descending">
      <ColumnDef objectType="ChangeItemRevision" propertyName="fnd0InProgress" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="awp0Item_item_id" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="object_type" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="item_revision_id" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="object_name" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="CMaturity" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="creation_date" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="last_mod_date" width="200"/>
      <ColumnDef objectType="ChangeItemRevision" propertyName="cm0Requestor" width="200"/>
    </ColumnConfig>
  </ClientScope>
</Client>
```

Configuring tables

Execute import_uiconfig utility

```

File=<File path>      Path of the Import File.
D:\TCUADEU\Siemens\Teamcenter10_1_7\bin>import_uiconfig.exe -u=infodba -p=infodba
a -g=dba -file=D:\TCUADEU\uiconfig\export.xml
Log File location : C:\Users\SMM914~1.MYT\AppData\Local\Temp\import_uiconfig_06
02-2018-14-39.log
$successfully processed the client scope fnd0Inbox
$successfully processed the client scope fnd0surrogatescope
$successfully processed the client scope Rv1xRTRelations
$successfully processed the client scope S2clAllQuestions
$successfully processed the client scope S2clAnsweredQuestions
$successfully processed the client scope S2clSocialCollaborationTrending
$successfully processed the client scope S2clUnAnsweredQuestions
$successfully processed the client scope Sub0SubscriptionsSettings
$successfully processed the client scope Arm0XRTRequirementViewer
$successfully processed the client scope awb0ContentTable.Awb0SavedBookmark
$successfully processed the client scope awb0ContentTable.Design Revision
$successfully processed the client scope awb0ContentTable.ItemRevision
$successfully processed the client scope awb0ContentTable.Paragraph Revision
$successfully processed the client scope awb0ContentTable.Part Revision
$successfully processed the client scope awb0ContentTable.Requirement Revision
$successfully processed the client scope awb0ContentTable.RequirementSpec Revision
$successfully processed the client scope Awb0OccurrenceManagement
$successfully processed the client scope Awp0AdvancedSearch
$successfully processed the client scope Awp0AllReports
$successfully processed the client scope Awp0CommandsWithConditions
$successfully processed the client scope Awp0Error
$successfully processed the client scope Awp0GlobalCommands
$successfully processed the client scope Awp0GlobalStableCommands
$successfully processed the client scope Awp0MyFavorites
$successfully processed the client scope Awp0MyProject
$successfully processed the client scope Awp0MyRecent
$successfully processed the client scope Awp0MyReports
$successfully processed the client scope Awp0ObjectNavigation
$successfully processed the client scope Awp0PdfPane
$successfully processed the client scope Awp0PrintOuts
$successfully processed the client scope Awp0RecentSearch
$successfully processed the client scope Awp0ReportBuilder
$successfully processed the client scope Awp0SavedSearch
$successfully processed the client scope Awp0SearchResults
$successfully processed the client scope Awp0ShowObject
$successfully processed the client scope Awp0ShowPdfFile
$successfully processed the client scope Awp0ShowTcGateway
$successfully processed the client scope Awp0ShowViewerAdmin
$successfully processed the client scope Awp0TasksToTrack
$successfully processed the client scope Awp0TeamTasks
$successfully processed the client scope Awp0XrtObjectSet
$successfully processed the client scope CmiApprovedChanges
$successfully processed the client scope CmiClosedChanges
$successfully processed the client scope CmiDisapprovedChanges
$successfully processed the client scope CmiInvestigateChanges
$successfully processed the client scope CmiMyChanges
$successfully processed the client scope CmiMyOpenChanges
$successfully processed the client scope CmiReportProblems

```

New Column added as shown below

Sanjoy Mondal (smm914024)-Engineering/Designer-Latest Working

Select: Owner Select: Category

← Changes

All Submitted Saved Approved Disapproved

4 results found for "All" >

IN PROCESS	ID	TYPE	REVISION	SYNOPSIS	MATURITY	DATE
False	17JM000009	ERC DML REVISION	A	Sanjoy Test	Complete	05-Feb
False	17JM000000_02	TASK REVISION	A	sanjoy task	Complete	05-Feb
False	17JM000008	ERC DML REVISION	A	Test Sanjoy	Complete	05-Feb
False	17JM000000_01	TASK REVISION	A	Test Task	Complete	05-Feb

- 1. Active Workspace development environment**
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Examples:-

- 1. Add Theme**
- 2. Create a Command**
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- 4. Create a new Pages**

The Active Workspace development environment consists of three important directories. Not all directories present in the environment are shown.

1. stage

This is the base directory for Active Workspace customization. It is located in **TC_ROOT\aws2**. This is where the environment initialization script (**initEnv**) is located.

2. components

This directory contains the OOTB source for Active Workspace, and is required when building the web application. Optional components, like **darsi** and **gateway** for example, may also be found here.

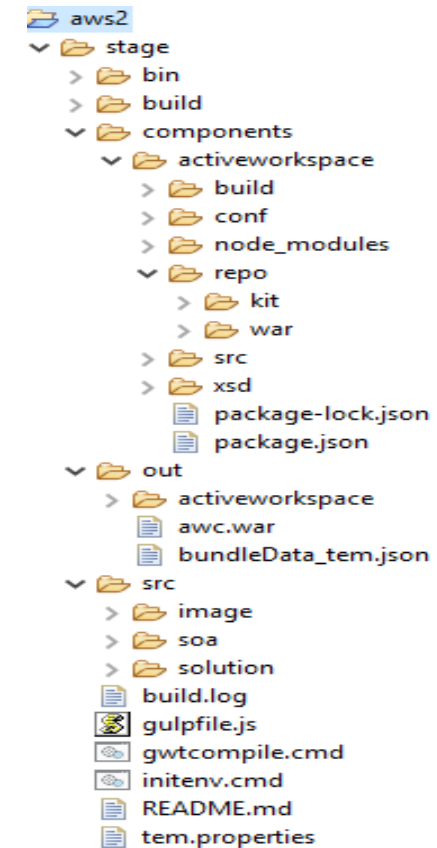
You can use the declarative source definitions in the **activeworkspace\repo\kit** directory for reference; all of the installed OOTB kits are located here in their respective directories.

3. out

This is where the build file can be found for deployment: **awc.war** for J2EE, and **awc.zip** for IIS. If the Gateway Client and the UI Builder are installed, you may find their working directories here as well.

4. src

The files and directories located here are for your customizations. They are pre-populated with actual source code and resources that are built along with the contents of the components directory.



The Active Workspace development scripts

To create custom Active Workspace components, you must work within an environment configured for Active Workspace development. Various scripts have been provided to assist in the development of component modules. Unless otherwise stated, all scripts must be run from the **stage** directory.

initEnv

Open a command prompt, and then run this script to configure the environment variables necessary for Active Workspace development. This only needs to be run once - when first opening a command prompt, before doing any other development work.

gwtCompile

This script compiles and builds the **awc** web application. By default, this compiles and builds the Active Workspace web application file: **awc.war** for J2EE or **awc.zip** for IIS.

This must be run from the **stage** directory.

generateModule

This script assists you in creating all the boilerplate files and directories for individual components. This script can be run from anywhere, it will place its artifacts in their proper locations. The **generateModule** script does not require any arguments. If you do not provide any, it will prompt you to enter the information it requires.

Your custom module must follow the declarative file and directory structure. Following is an example of a module with several customizations

Using generateModule to create boilerplate definitions

Creating your own custom declarative definitions for Active Workspace requires a container so the build scripts can locate your files. This container is a **kit.json** file. If your kit file is located anywhere under the **stage\src** directory, it will be found by the build scripts. Each kit file contains a list of modules, solution definitions, and other dependency information. The system will automatically find any registered modules under the **stage\src** directory.

Typically, you would create one kit that contains several modules, each of which contains your declarative definitions. Active Workspace ships with a single kit already in place, the solution kit, located in the **stage\src\solution** directory.

As long as you are working in the development environment (set up using `initenv`), you can run **generateModule**. You do not have to be in any particular directory. The utility will prompt you for the input it requires, and it validates many of the choices you must make against existing options.

Using generateModule to create boilerplate definitions

The utility currently assists you with the creation of the following declarative components.

- **module**

Create your own module to contain related declarative configuration files. The utility will create a directory for each of your custom modules in the `stage\src` directory and create the various module files.

*If you do not explicitly create your own module, **mysamplemodule** will be created for you.*

- **type**

Use the declarative framework to assign an icon to an object type. The utility will create **aliasRegistry.json** in the module directory. Using alias registry allows you to register an icon from the `stage\src\image` directory to a given object type or types.

- **theme**

Create a new theme that you can modify. The utility will create **ui-{themeName}.scss** in the module's `src` directory. This new scss file will be configured with your color choice, but can be modified if desired.

- **command**

Define a command. The utility creates a command definition in **commandsViewModel.json** in the module directory.

Tip

Siemens PLM Software recommends installing the **Gateway Client** and the **UI Builder** to make changes to commands. As part of **Active Architect**, these provide you with the **Command Builder** and **Panel Builder** locations for easier command editing and creation.

Using generateModule to create boilerplate definitions

- **location**

Create a new location to organize your sublocations. The utility creates a location definition in **states.json** in the module directory.

- **subLocation**

Create a new sublocation. The utility creates a sublocation definition in **states.json** in the module directory. Sublocations must be assigned to an existing location.

- **workspace**

Create the server-side definitions for a new workspace. The utility creates a workspace definition file in the **solution** directory, and adds an entry in the main **kit.json**. In the module directory, it creates an entry in the messages file.

```
C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage>generateModule
[Info] Select a type to generate. Available types are:

command: Generates a zero compile command.

kit: Generates a kit.json.
A kit.json is a combination of modules which will be included in the site build

location: Generates a location.
A location groups sublocations together within Active Workspace.
Examples are "Inbox" and "Search"

module: Generates a module.
A module is a build configuration that determines which code to include and how to process it.

state: Generates a state. A state is a page within a solution. The page's display is a declarative view and view model

subLocation: Generates a subLocation.
sublocation is a page which can be accessed in Active Workspace. The standard view consists of a primary and secondary workarea.
Examples are "My Tasks" within "Inbox" location and "Results" within "Search" location

theme: Generates a theme.
A theme can change the colors that are used in your site. The default themes are Light and Dark

type: Generates a type.
This will allow you to define a custom type icon for a type.

workinstrView: Generates a zero compile workinstr gallery view within EWI solution.

workspace: Generates a workspace.

Enter type to generate: module
[13:13:34] info: Starting the module generator

Give the module a name. This is a unique identifier used by the build system.

Module name: awctrainingmodule
[13:14:47] info: Created module awctrainingmodule in C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage/src/awctrainingmodule
[13:14:47] info: Selected kit tc-aw-solution for modification.
[13:14:48] info: Added module awctrainingmodule to kit tc-aw-solution

C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage>
```

> This PC > OS (C:) > Siemens > AwcTrng > Teamcenter12 > aws2 > stage > src				
	Name	Date modified	Type	Size
	awctrainingmodule	13-09-2021 13:14	File folder	
	image	31-07-2019 02:19	File folder	
	soa	31-07-2019 02:19	File folder	
	solution	31-07-2019 02:19	File folder	
	.eslintrc.js	31-07-2019 02:19	JavaScript File	2 KB
	build.log	02-09-2021 17:27	Text Document	1 KB

kin

ing

Variables defined in SCSS files control all of the colors for Active Workspace. When you change the SCSS file, it propagates throughout the interface. Changing individual CSS files is not supported.

```
C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage>generateModule
[Info] Select a type to generate. Available types are:

command: Generates a zero compile command.

kit: Generates a kit.json.
A kit.json is a combination of modules which will be included in the site build

location: Generates a location.
A location groups sublocations together within Active Workspace.
Examples are "Inbox" and "Search"

module: Generates a module.
A module is a build configuration that determines which code to include and how to process it.

state: Generates a state. A state is a page within a solution. The page's display is a declarative view and view model

sublocation: Generates a sublocation.
sublocation is a page which can be accessed in Active Workspace. The standard view consists of a primary and secondary
orkarea.
Examples are "My Tasks" within "Inbox" location and "Results" within "Search" location

theme: Generates a theme.
A theme can change the colors that are used in your site. The default themes are Light and Dark

type: Generates a type.
This will allow you to define a custom type icon for a type.

workinstrView: Generates a zero compile workinstr gallery view within EWI solution.

workspace: Generates a workspace.

Enter type to generate: theme
[13:24:10] info: Starting the theme generator
[13:24:10] info: Selected module awctrainingmodule for modification.

Give the theme a name. This will be used as the title of your theme and scss file.
Theme name: myGreenTheme

Give the theme a base color. This will be used as the primary color of your theme with different shades as secondary co
ors.
This can be a CSS color name such as red or a hex color such as #FF0000
Color: green
[13:24:48] info: Created ui-myGreenTheme.scss
[13:24:48] info: Created theme myGreenTheme in module awctrainingmodule
```

Various files and directories are created or updated, including a new command to switch to your theme, but there are two main files of interest:

- **theme.scss**

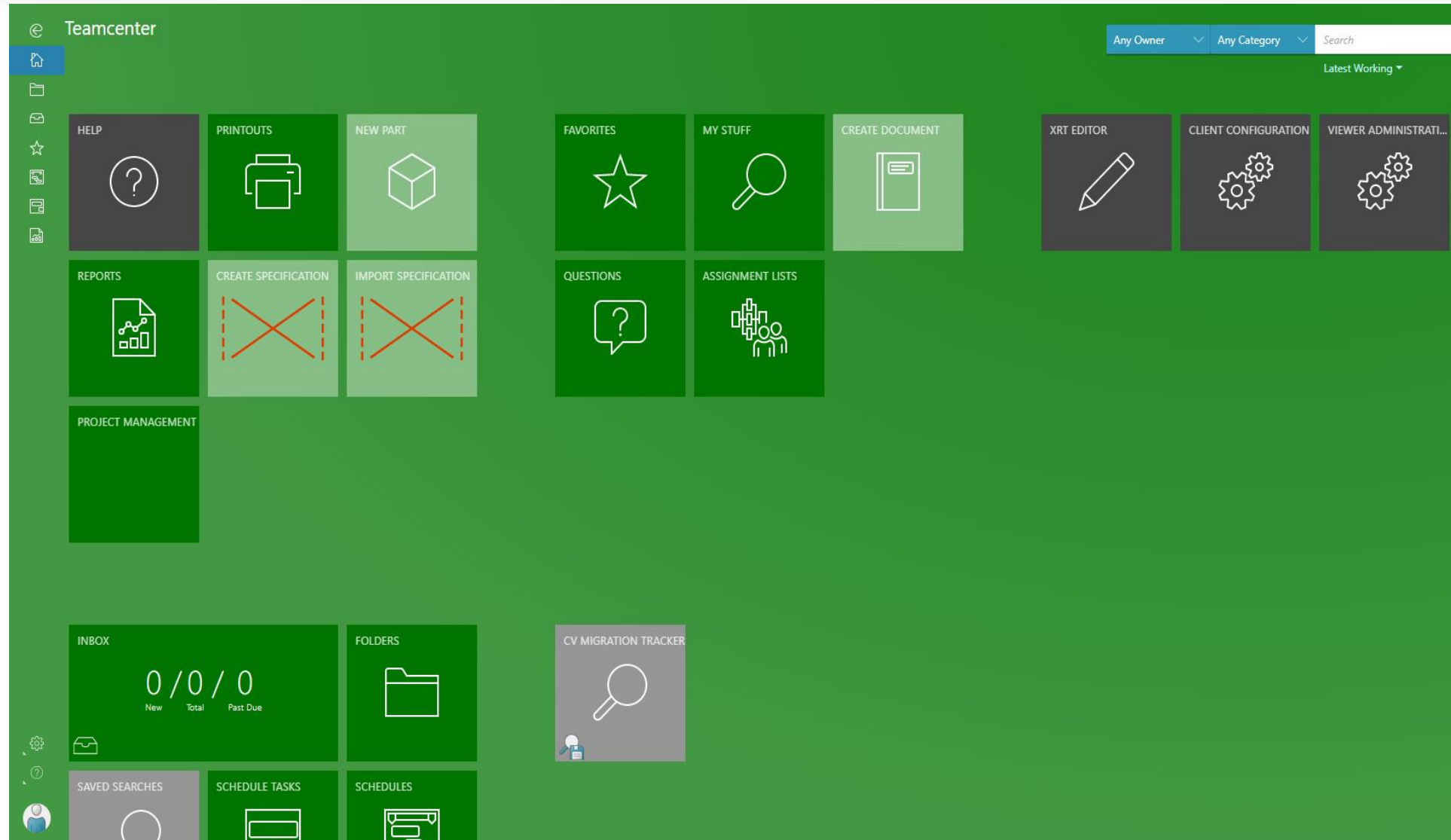
Your theme file is located in the module's **src** directory. In this example, **ui-myGreenTheme.scss**. This file contains the base color definitions, and can be modified to suit your needs. See the following example.

- **messages.json**

Your messages file is located in the module's **src\i18n** directory. In this example, **awctrainingmoduleMessages.json**. This file contains translations for your module, including the command title, **myGreenThemeTitle**. In this example, change the title to something more easily understood by the user.

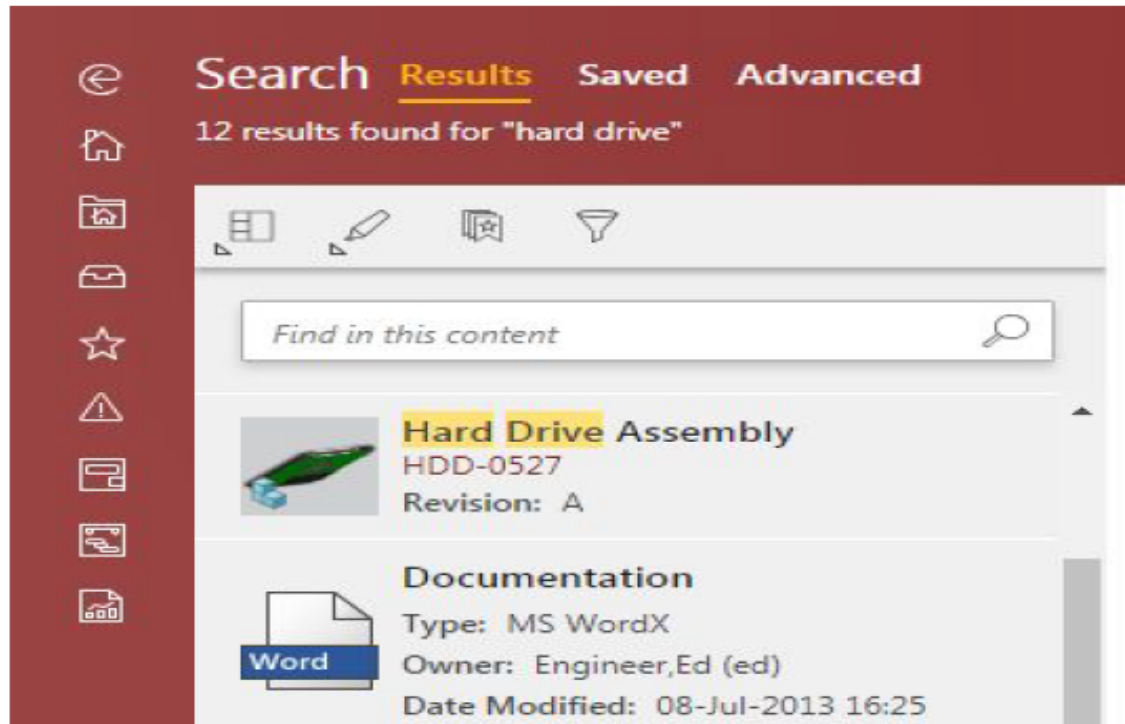
```
"myGreenThemeTitle": "Green"
```

Build and deploy the awc application.



Overriding a default color

You feel the red theme doesn't go far enough, search results are highlighted in yellow, but you want red. In this example, you change the color of Active Workspace's highlighting from yellow to red.



1. Examine the OOTB theme scss files to find the variable you want to override. In this case it is the highlighted background color.

The provided theme files can be found under the **stage\repo** directory. Custom themes are based on the **_ui-theme.scss** and **_ui-theme-settings.scss**.

Overriding a default color

			▼ ⓘ "theme".scss
	_clsTheme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:20 Size: 799 bytes
	_kanbanInterfaceTheme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 2.12 KB
	ui-__theme-name___.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 1.48 KB
	_afx-base-themeColors.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 1.76 KB
	_afx-theme-settings.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 15.3 KB
	_afx-theme-vars.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 7.05 KB
	_afx-widgets-theme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 28.0 KB
	_ui-base-themeColors.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 712 bytes
	_ui-theme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 90.7 KB
	_ui-theme-settings.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 7.21 KB
	_ui-theme-vars.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 3.92 KB
	ui-darkTheme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 3.33 KB
	ui-lightTheme.scss C:\Siemens\AwcTrng\Teamcenter12\aws2\stage\co...	Type: SCSS File	Date modified: 31-07-2019 02:19 Size: 943 bytes

2. Copy the line defining the highlighted background color from the ui theme settings scss.

```
$aw_highlighted_background_color: $AW_SiemensAccentYellow5;
```

3. Paste into your custom theme scss file after the theme settings import, but before the ui theme import.

```
@import 'ui-theme-settings';
$aw_highlighted_background_color: $AW_SiemensAccentYellow5;
@import 'ui-theme';
```

4. Change the color of the highlighting. You could specify any standard color or hexadecimal color code. In this example, use one of the defined red highlight colors.

```
@import 'ui-theme-settings';
$aw_highlighted_background_color: $AW_SiemensAccentRed5;
@import 'ui-theme';
```

5. Build and deploy the awc application.
6. Test your new highlight color.

Declarative commands require a native module. Its location is determined by which **uiAnchor** you use. In this case, the right wall command bar.

Step 1: Use the **generateModule** script to create the command. In this example, you create **myCommand**, choosing an existing icon and placing it on the right wall command bar.

```
C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage>generateModule
[Info] Select a type to generate. Available types are:

command: Generates a zero compile command.

kit: Generates a kit.json.
A kit.json is a combination of modules which will be included in the site build

location: Generates a location.
A location groups sublocations together within Active Workspace.
Examples are "Inbox" and "Search"

module: Generates a module.
A module is a build configuration that determines which code to include and how to process it.

state: Generates a state. A state is a page within a solution. The page's display is a declarative view and view model

subLocation: Generates a subLocation.
sublocation is a page which can be accessed in Active Workspace. The standard view consists of a primary and secondary workarea.
Examples are "My Tasks" within "Inbox" location and "Results" within "Search" location

theme: Generates a theme.
A theme can change the colors that are used in your site. The default themes are Light and Dark

type: Generates a type.
This will allow you to define a custom type icon for a type.

workinstrView: Generates a zero compile workinstr gallery view within EWI solution.

workspace: Generates a workspace.

Enter type to generate: command
[14:42:34] info: Starting the command generator
[14:42:34] info: Selected module awctrainingmodule for modification.

Give the command a name. This is a unique identifier referenced by handlers and placements.
Command name: myCommand
```

```
Number,cmdGenerateTransactions,cmdGeometric,cmdGeometricCrossSection,cmdGeometricMeasure,cmdGeometricQuery,cmdGeometricVolume,cmdGrantModificationRights,cmdGrid,cmdGroupExpression,cmdGuide,cmdHalfFilledStar,cmdHideGraphOverview,cmdHighlight,cmdHome,cmdHomeLHN,cmdHorizontalSplit,cmdImage,cmdImage,cmdImport,cmdImportDeclaration,cmdImportExport,cmdImportExportExcel,cmdImportExportWord,cmdInbox,cmdIncludeVariants,cmdIncomingPartial,cmdIndexedModel,cmdInitiateSuspect,cmdInsert,cmdInsertImage,cmdInsertOLE,cmdJoinProcesses,cmdJoinProcesses,cmdLargeImageView,cmdLaunchSimulationTool,cmdLaunchWorkflow,cmdLayoutGroup,cmdLayoutHierarchy,cmdLayoutSequence,cmdLicense,cmdLinktoNX,cmdListSummaryView,cmdListView,cmdMail,cmdManage,cmdManageDataCollectionSignatures,cmdManageDimensions,cmdManageWorkPackage,cmdManualValidation,cmdMap,cmdMarkup,cmdMassAssign,cmdMatrixView,cmdMeasurementManagement,cmdMeasureTape,cmdMenu,cmdMergeBranches,cmdMissingImage,cmdModelViewPalette,cmdMore,cmdMoveDown,cmdMoveHere,cmdMoveTo,cmdMoveToLeftList,cmdMoveToRightList,cmdMoveToSection,cmdMoveUp,cmdMultipleCreate,cmdMultiSelect,cmdNavigateAssociated,cmdNestedOn,cmdNestedOnOff,cmdNewBuildElement,cmdNoteProperties,cmdOpen,cmdOpenInbox,cmdOpenInHostApplication,cmdOpenInNewTab,cmdOpenInNx,cmdOpenPackages,cmdOpenSharedEffectivity,cmdOppositeSelection,cmdOrientBack,cmdOrientBottom,cmdOrientFront,cmdOrientISO,cmdOrientLeft,cmdOrientRight,cmdOrientTop,cmdOrientTrimetric,cmdOutgoingPartial,cmdOverrideComplianceResult,cmdPack,cmdPan,cmdPanelBuilder,cmdParent,cmdParentPartial,cmdPaIcon name: laneXy,cmdPlaneXz,cmdPlaneYz,cmdPlanNegativeXz,cmdPlay,cmdPMI,cmdPoint,cmdPorts,cmdPortsPartial,cmdPostEvent,cmdPrint,cmdProductionProgram,cmdPromote,cmdPromoteTask,cmdPropagateSelection,cmdProximity,cmdProxyBid,cmdPublish,cmdQuickLink,cmdRadioSelected,cmdRadioUnselected,cmdRecipe,cmdRecordPartMovement,cmdRecordUtilization,cmdRedo,cmdReflection,cmdRefresh,cmdRefreshFilter,cmdRegenerateBarcode,cmdReject,cmdRelatedObjects,cmdRelateProgramObjects,cmdRelease,cmdRemoteUpload,cmdRemove,cmdRemoveAll,cmdRemoveAssignment,cmdRemoveAttribute,cmdRemoveFromStudy,cmdRemoveFromValidationContract,cmdRemoveLocation,cmdRemoveVariants,cmdReplace,cmdReplaceFile,cmdReplaceLatestRevision,cmdReply,cmdReports,cmdRequestSupplierDeclaration,cmdRescheduleDataRequirementItem,cmdReset,cmdResetForm,cmdResolvePartMovement,cmdRetailImageTemplate,cmdReuseDocument,cmdRevertBranchToLatest,cmdRevertToDiscipline,cmdReviseToSolution,cmdRotate,cmdSave,cmdSaveAndExit,cmdSaveAs,cmdSaveAsDiagram,cmdSaveCost,cmdSaveDiagram,cmdSaveSearch,cmdSaveWhatIf,cmdSaveWorkingContext,cmdSchedules,cmdScheduleTasks,cmdSearch,cmdSearchLatestData,cmdSearchReport,cmdSelectFeatures,cmdSelectParts,cmdSendToSourcingManager,cmdSetActiveBranch,cmdSetAnchor,cmdSetAsContextSeason,cmdSetAsPredominantColor,cmdSetBaselineBranch,cmdSetLineage,cmdSetMeasurableAttributeInput,cmdSetMeasurableAttributeInputOutput,cmdSetMeasurableAttributeNone,cmdSetMeasurableAttributeOutput,cmdSetPrimary,cmdSetPrimary,cmdSetStepStatus,cmdSetTemporaryNotifier,cmdSettings,cmdSettingsInProgress,cmdShadeMode,cmdShadow,cmdShapeSearch,cmdShapeSearchFilter,cmdShare,cmdShareCheckout,cmdShiftDate,cmdShow,cmdShowAggregatedPartsAndTools,cmdShowAllPorts,cmdShowAllResultsIn3dViewer,cmdShowAssociated,cmdShowAttachments,cmdShowCapsAndCutLines,cmdShowChildren,cmdShowCommentaryPanel,cmdShowConnectedPorts,cmdShowConnection,cmdShowDigitalAssets,cmdShowFiltered,cmdShowGraphOverview,cmdShowGrid,cmdShowImpactOfChange,cmdShowIncomingRelations,cmdShowInfoPanel,cmdShowInterfaces,cmdShowMarkupPanel,cmdShowOnlyUpToDate,cmdShowOutgoingRelations,cmdShowParentBranch,cmdShowRelationsControls,cmdShowTable,cmdShowUntraced,cmdShowUnusedAttributes,cmdSignatureCertificate,cmdSignout,cmdSignOut,cmdSimToolProgressMonitor,cmdSort,cmdSortAscending,cmdSortDecending,cmdSourcingManager,cmdStandIn,cmdStationPartMode,cmdStop,cmdStructureNavigate,cmdSubmitPartMovement,cmdSubmitToWorkflow,cmdSubstanceCategorize,cmdSurface,cmdSwap,cmdSystemSelection,cmdTableSummaryView,cmdTableView,cmdThumbsUp,cmdThumbsUpFilled,cmdTime,cmdToggleBuildCondition,cmdToggleCompartment,cmdToggleLabel,cmdTracelinkPartial,cmdTransfer,cmdTravelerReport,cmdTreeSummary,cmdTreeView,cmdTrihedron,cmdUndo,cmdUngroupExpression,cmdUnlink,cmdUnlinkfromNX,cmdUnmap,cmdUnpack,cmdUnpin,cmdUnpostEvent,cmdUnsetLineage,cmdUnsubmitResponse,cmdUp,cmdUpdateAssignedRevision,cmdUpdateDeliverableSequence,cmdUpdateDependantCosts,cmdUpdateSchedule,cmdUpload,cmdUpToDate,cmdUser,cmdUserAvailability,cmdUseTransparencyOff,cmdUseTransparencyOn,cmdValidate,cmdValidateVariantConfiguration,cmdValidateVariantExpression,cmdVariantOptions,cmdVertex,cmdVerticalSplit,cmdView,cmdViewConvenience,cmdViewFlat,cmdViewSingleBranch,cmdViewSingleLevel,cmdWalk,cmdWhatIfAnalysis,cmdWorkInProgressLimit,cmdWrapText,cmdXyPlanar,cmdXzPlanar,cmdYzPlanar,cmdZoom,cmdZxPlanar
```

Icon name: cmdPlay

Choose a placement for this command.

This placement should match the anchor of the command bar you want this command to appear in.

Common placements are:

```
aw_rightWall
aw_oneStep
aw_primaryWorkArea
aw_3dViewer
aw_relationsViewer
aw_footer
aw_universalViewer
aw_toolsAndInfo
aw_ganttViewer
aw_graph_node
```

```
Icon name: cmdPlay

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aw_ganttViewer
aw_graph_node

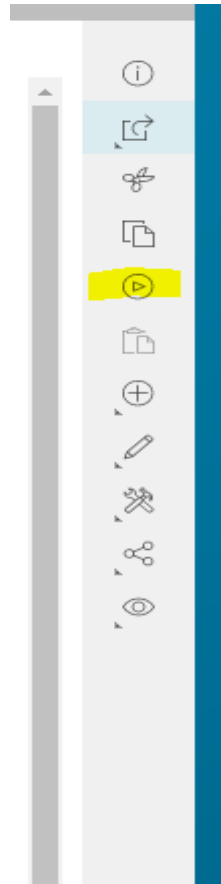
Placement name: aw_rightWall
[14:47:38] info: Created myCommandService.js
[14:47:38] info: Created command myCommand in module awctrainingmodule

C:\Siemens\AwcTrng\TEAMCE~1\aws2\stage>
```

```

v awctrainingmodule
  v src
    v i18n
      awctrainingmoduleMessages.json
    v js
      myCommandService.js
      ui-myGreenTheme.scss
      commandsViewModel.json
      module.json
```

Step 2: Build the application, deploy, and test.



What is the declarative UI?

A capability provided by the Active Workspace framework that allows for a concise and codeless definition of UI view content, layout, routing, and behavior. Actions, messages, service calls and their inputs and outputs can be mapped and described using **HTML** and **JSON**. It provides an *abstracted* means of defining the client UI and its behaviors; the underlying implementation is hidden.

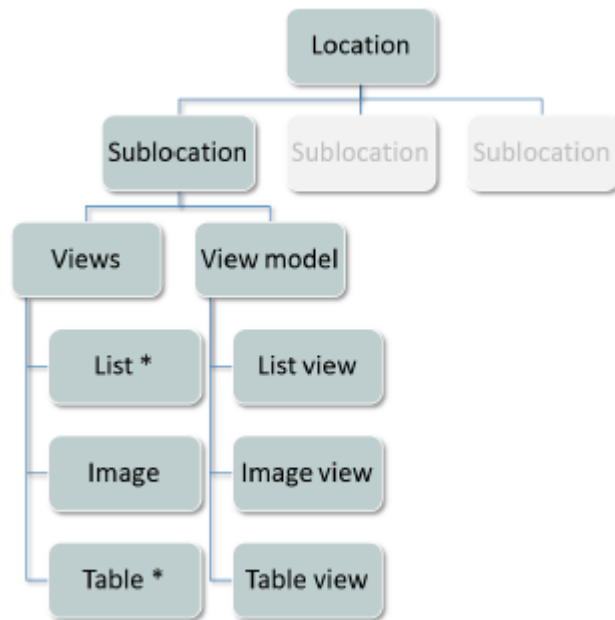
Why a declarative UI?

There are many reasons to use an abstracted, declarative user interface. It provides increased:

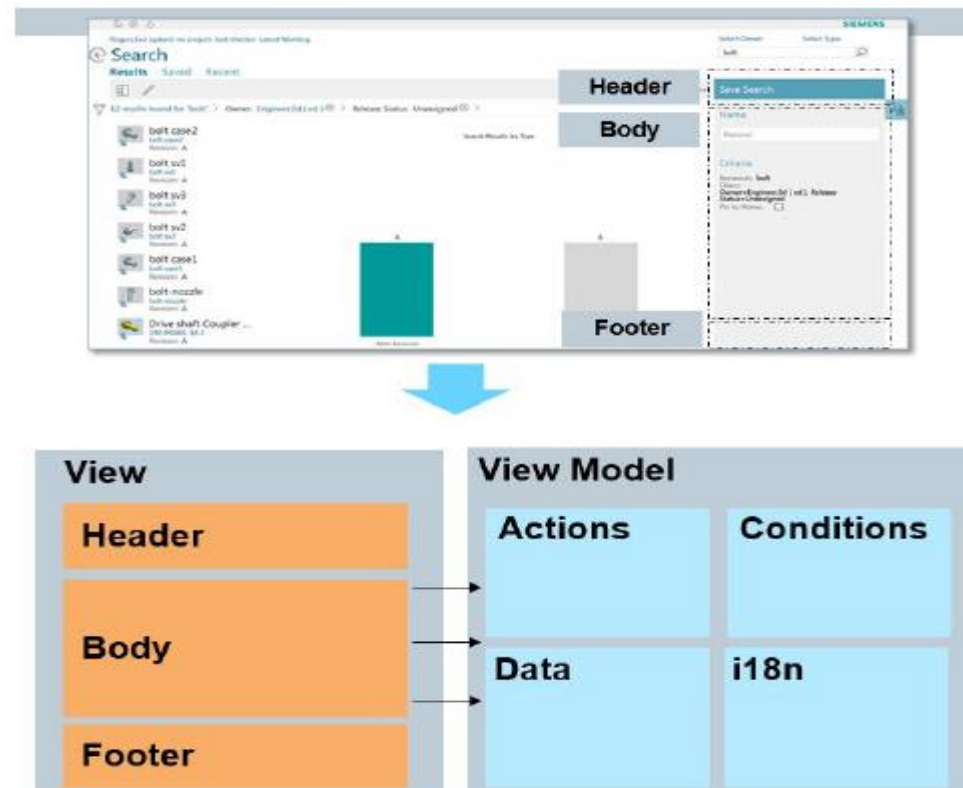
- 1. Performance** - Reducing the number of lines of code decreases load times.
- 2. Efficiency** - Enables faster iteration and reduces the need to develop and maintain code. A simple declarative view definition allows UX specialists to author the desired UI, while working with an application developer to wire up the view model based on the intent.
- 3. Sustainability** - Since the Active Workspace declarative elements are abstracted, the underlying web technology can evolve with minimal to no effect on existing layouts.
- 4. Consistency** - Each page and panel element is built from basic, modular UI elements instead of custom pieces.

UI architecture

The Active Workspace UI consists mainly of sublocations grouped by locations. Each sublocation is defined by views, which rely on view models for their functionality.



The declarative definition of the UI has two main components, the view and the view model.



THANK YOU